10 $f(x) = 4\sin x°$

(a) Find f(23)
Give your answer correct to 3 significant figures.

$f(23) = 4 \sin 23° = 1.562924514...$

1 mark
Correct value; use degrees.

1.56

(1)

$g(x) = 2x - 3$

(b) Find fg(34)
Give your answer correct to 3 significant figures.

$g(34) = 2 \times 34 - 3 = 65$

$fg(34) = f(65) = 4 \sin 65°$

$= 3.625231148...$

1 mark
Apply g first, then apply f to its output.

1 mark
Correct composite value.

3.63

(2)

$h(x) = (x + 4)^2$

Ivan needs to solve the following equation $h(x) = 25$

He writes

$(x + 4)^2 = 25$
$x + 4 = 5$
$x = 1$

This is not fully correct.

(c) Explain why.

$x + 4 = 5$ or -5

1 mark
Both square roots are needed.

So $x = 1$ or -9.

Ivan missed the negative square root.

(1)

(Total for Question 10 is 4 marks)